

Kubernetes

Assignment 1:

Tasks To Be Performed:

1. Deploy a Kubernetes cluster for 3 nodes 2.

Create a NGINX deployment of 3 replicas

\$ minikube start --nodes=2

* minikube v1.38.1 on Microsoft Windows 11 Home 25H2

* Automatically selected the docker driver

! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.

* Using Docker Desktop driver with root privileges

* Starting "minikube" primary control-plane node in "minikube" cluster

* Pulling base image v0.0.50 ...

* Configuring CNI (Container Networking Interface) ...

* Verifying Kubernetes components...

- Using image gcr.io/k8s-minikube/storage-provisioner:v5

* Enabled addons: storage-provisioner, default-storageclass

* Starting "minikube-m02" worker node in "minikube" cluster

* Pulling base image v0.0.50 ...

* Found network options:

- NO_PROXY=192.168.49.2

- NO_PROXY=192.168.49.2

- env NO_PROXY=192.168.49.2

* Verifying Kubernetes components...

! C:\Users\\$\$\$\$\bin\kubectl.exe is version 1.30.0, which may have incompatibilities with Kubernetes 1.35.1.

- Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'

* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

\$ minikube node add

* Adding node m03 to cluster minikube as [worker]

* Starting "minikube-m03" worker node in "minikube" cluster

* Pulling base image v0.0.50 ...

* Verifying Kubernetes components...

* Successfully added m03 to minikube!

\$ kubectl get nodes

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	9m38s	v1.35.1
minikube-m02	Ready	<none>	9m13s	v1.35.1
minikube-m03	NotReady	<none>	7s	v1.35.1 → Not ready due to insufficient capacity

\$ kubectl create deployment nginx --image=nginx --replicas=3

deployment.apps/nginx created

\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
nginx-56c45fd5ff-h4wtq	1/1	Running	0	84s
nginx-56c45fd5ff-m4slf	1/1	Running	0	84s
nginx-56c45fd5ff-sk999	1/1	Running	0	84s

Assignment 2:

Tasks To Be Performed:

1. Use the previous deployment

2. Create a service of type NodePort for NGINX deployment

3. Check the NodePort service on a browser to verify

\$ kubectl get svc

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	32m

\$ kubectl expose deployment nginx --type=NodePort --port=80

service/nginx exposed

\$ kubectl get svc

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	33m
nginx	NodePort	10.111.182.234	<none>	80:31792/TCP	8s

\$ minikube service nginx

NAMESPACE	NAME	TARGET PORT	URL	
default	nginx	80	http://192.168.49.2:31792	

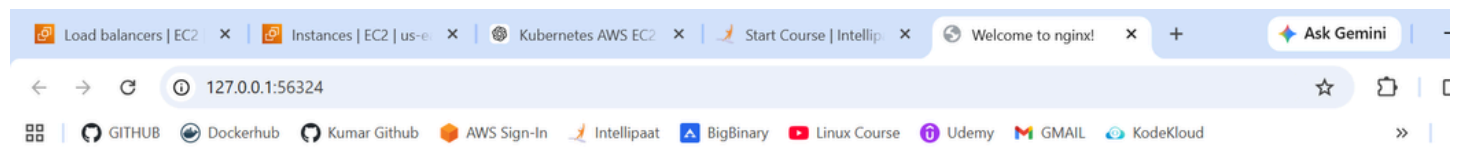
* Starting tunnel for service nginx.

NAMESPACE	NAME	TARGET PORT	URL	
default	nginx		http://127.0.0.1:56324	

* Opening service default/nginx in default browser...

! Because you are using a Docker driver on windows, the terminal needs to be open to run it.

* Stopping tunnel for service nginx.



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Assignment 3:

Tasks To Be Performed:

1. Use the previous deployment
2. Change the replicas to 5 for the deployment

\$ kubectl get deployment

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
------	-------	------------	-----------	-----

nginx	3/3	3	3	2m4s
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\$ kubectl scale deployment nginx --replicas=5

deployment.apps/nginx scaled

\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

nginx-56c45fd5ff-57w9n	1/1	Running	0	4s
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nginx-56c45fd5ff-h4wtq	1/1	Running	0	2m17s
------------------------	-----	---------	---	-------

```

nginx-56c45fd5ff-hdgrm 1/1 Running 0 4s
nginx-56c45fd5ff-m4slf 1/1 Running 0 2m17s
nginx-56c45fd5ff-sk999 1/1 Running 0 2m17s

```

Assignment 4:

Tasks To Be Performed:

1. Use the previous deployment

2. Change the service type to ClusterIP

```
$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	33m
nginx	NodePort	10.111.182.234	<none>	80:31792/TCP	8s

```
$ kubectl patch svc nginx -p '{"spec":{"type":"ClusterIP"}}'
```

```
service/nginx patched
```

```
$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	37m
nginx	ClusterIP	10.111.182.234	<none>	80/TCP	3m26s

Assignment 5:

Tasks To Be Performed:

1. Use the previous deployment

2. Deploy an NGINX deployment of 3 replicas

3. Create an NGINX service of type ClusterIP

4. Create an ingress service/ Apache to Apache service/ NGINX to NGINX service

```
$ kubectl create namespace ingress-nginx
```

```
namespace/ingress-nginx created
```

```
$ kubectl apply -f https://raw.githubusercontent.com/kubernetes/ingress-nginx/main/deploy/static/provider/cloud/deploy.yaml
```

Warning: resource namespaces/ingress-nginx is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.

namespace/ingress-nginx configured

serviceaccount/ingress-nginx created

serviceaccount/ingress-nginx-admission created

role.rbac.authorization.k8s.io/ingress-nginx created

role.rbac.authorization.k8s.io/ingress-nginx-admission created

clusterrole.rbac.authorization.k8s.io/ingress-nginx unchanged

clusterrole.rbac.authorization.k8s.io/ingress-nginx-admission unchanged

rolebinding.rbac.authorization.k8s.io/ingress-nginx created

rolebinding.rbac.authorization.k8s.io/ingress-nginx-admission created

clusterrolebinding.rbac.authorization.k8s.io/ingress-nginx unchanged

clusterrolebinding.rbac.authorization.k8s.io/ingress-nginx-admission unchanged

configmap/ingress-nginx-controller created

service/ingress-nginx-controller created

service/ingress-nginx-controller-admission created

deployment.apps/ingress-nginx-controller created

job.batch/ingress-nginx-admission-create created

job.batch/ingress-nginx-admission-patch created

ingressclass.networking.k8s.io/nginx unchanged

validatingwebhookconfiguration.admissionregistration.k8s.io/ingress-nginx-admission configured

\$ kubectl get pods -n ingress-nginx

NAME	READY	STATUS	RESTARTS	AGE
ingress-nginx-controller-7d65c586d6-pcwf6	1/1	Running	0	19s

\$ kubectl get svc -n ingress-nginx

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
------	------	------------	-------------	---------	-----

ingress-nginx-controller	LoadBalancer	10.110.241.223	<pending>	80:32762/TCP,443:30290/TCP	26s
ingress-nginx-controller-admission	ClusterIP	10.96.151.15	<none>	443/TCP	26s

Load balancer:

\$ minikube tunnel

* Tunnel successfully started

* NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

! Access to ports below 1024 may fail on Windows with OpenSSH clients older than v8.1. For more information, see: <https://minikube.sigs.k8s.io/docs/handbook/accessing/#access-to-ports-1024-on-windows-requires-root-permission>

* Starting tunnel for service ingress-nginx-controller.

! Access to ports below 1024 may fail on Windows with OpenSSH clients older than v8.1. For more information, see: <https://minikube.sigs.k8s.io/docs/handbook/accessing/#access-to-ports-1024-on-windows-requires-root-permission>

* Starting tunnel for service demo-ingress.

\$ kubectl get svc -n ingress-nginx

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
ingress-nginx-controller	LoadBalancer	10.110.241.223	127.0.0.1	80:32762/TCP,443:30290/TCP	85s
ingress-nginx-controller-admission	ClusterIP	10.96.151.15	<none>	443/TCP	85s

\$ kubectl get pod -n ingress-nginx

NAME	READY	STATUS	RESTARTS	AGE
ingress-nginx-controller-7d65c586d6-pcwf6	1/1	Running	0	2m3s

\$ kubectl create namespace demo-app

namespace/demo-app created

\$ kubectl get ns demo-app

NAME	STATUS	AGE
------	--------	-----

demo-app Active 41m

\$ vi nginx_code.yaml

\$ cat nginx_code.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx

namespace: demo-app

spec:

replicas: 2

selector:

matchLabels:

app: nginx

template:

metadata:

labels:

app: nginx

spec:

containers:

- name: nginx

image: hashicorp/http-echo

args:

- "-text=Nginx page"

apiVersion: v1

kind: Service

metadata:

name: nginx

namespace: demo-app

spec:

type: ClusterIP

selector:

app: nginx

ports:

- port: 80

targetPort: 5678

\$ kubectl apply -f nginx_code.yaml

deployment.apps/nginx created

service/nginx created

\$ vi apache_code.yaml

\$ cat apache_code.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: apache

namespace: demo-app

spec:

replicas: 2

selector:

matchLabels:

app: apache

template:

metadata:

labels:

app: apache

spec:

containers:

- name: apache

image: hashicorp/http-echo

args:

- "-text=Apache page"

apiVersion: v1

kind: Service

metadata:

name: apache

namespace: demo-app

spec:

type: ClusterIP

selector:

app: apache

ports:

- port: 80

targetPort: 5678

\$ kubectl apply -f apache_code.yaml

deployment.apps/apache created

service/apache created

```
$ kubectl get pods -n demo-app
```

NAME	READY	STATUS	RESTARTS	AGE
apache-59c484b95b-84c24	1/1	Running	0	9m46s
apache-59c484b95b-wh742	1/1	Running	0	9m46s
nginx-c4b6f588b-4rw2k	1/1	Running	0	15m
nginx-c4b6f588b-gcrrh	1/1	Running	0	15m

```
$ vi demo-ingress.yaml
```

```
$ cat demo-ingress.yaml
```

```
apiVersion: networking.k8s.io/v1
```

```
kind: Ingress
```

```
metadata:
```

```
  name: demo-ingress
```

```
  namespace: demo-app
```

```
spec:
```

```
  ingressClassName: nginx
```

```
  rules:
```

```
    - http:
```

```
      paths:
```

```
        - path: /nginx
```

```
          pathType: Prefix
```

```
      backend:
```

```
        service:
```

```
          name: nginx
```

```
          port:
```

```
            number: 80
```

```
- path: /apache

pathType: Prefix

backend:

  service:

    name: apache

    port:

      number: 80
```

\$ kubectl apply -f demo-ingress.yaml

ingress.networking.k8s.io/demo-ingress configured

\$ kubectl get ingress -n demo-app

NAME	CLASS	HOSTS	ADDRESS	PORTS	AGE
demo-ingress	nginx	*	127.0.0.1	80	10m

\$ kubectl get ns

NAME	STATUS	AGE
default	Active	31h
demo-app	Active	81m
ingress-nginx	Active	86m
kube-node-lease	Active	31h
kube-public	Active	31h
kube-system	Active	31h

\$ kubectl get ns ingress-nginx

NAME	STATUS	AGE
ingress-nginx	Active	86m

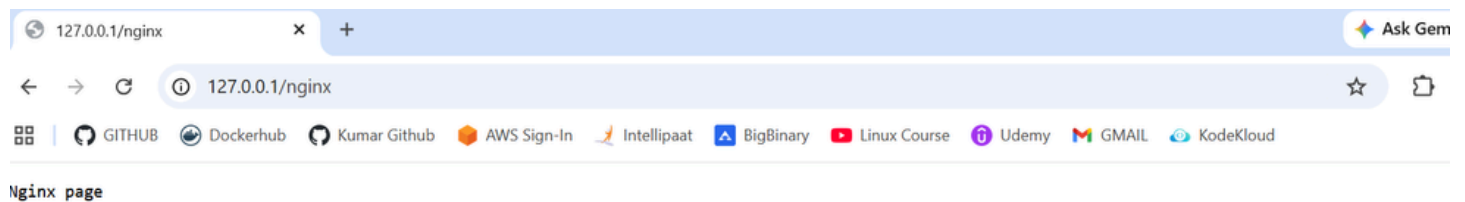
\$ kubectl get pods -n demo-app

NAME	READY	STATUS	RESTARTS	AGE
apache-59c484b95b-84c24	1/1	Running	0	62m

apache-59c484b95b-wh742	1/1	Running	0	62m
nginx-c4b6f588b-4rw2k	1/1	Running	0	67m
nginx-c4b6f588b-gcrrh	1/1	Running	0	67m
service-a-846968bc9c-lgh8r	1/1	Running	0	78m
service-a-846968bc9c-q94m2	1/1	Running	0	78m
service-b-5fd8bd5b68-dw5rz	1/1	Running	0	78m
service-b-5fd8bd5b68-fgqwd	1/1	Running	0	78m

\$ kubectl get nodes

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	31h	v1.35.1
minikube-m02	Ready	<none>	31h	v1.35.1
minikube-m03	Ready	<none>	10h	v1.35.1



Apache page